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Multimodal AI-Driven Framework for Real-Time Compliance Monitoring

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Abstract

This paper introduces an AI-powered solution for automated Health, Safety, and Environmental (HSE) compliance monitoring in dynamic industrial environments such as oil rigs and workshops. Our workflow leverages a sophisticated multimodal video analysis pipeline, integrated with Standard Operating Procedures (SOPs), to provide automated, real-time detection of operational violations. A critical advancement is the solution's optimized design for edge deployment, which guarantees robust, real-time, and highly accurate insights. This edge-centric workflow ensures scalability across various industrial applications, effectively mitigating safety risks, substantially reducing the burden of manual inspections, and providing a verifiable, SOP-compliant monitoring framework.

Introduction

Traditional real-time object detection frameworks such as You Only Look Once (YOLO) ([Redmon et.al, 2016](#)) have been widely used to detect personal protective equipment (PPE), and other safety gear on industrial sites. YOLO, in particular, is known for its speed and efficiency, making it ideal for applications where real-time decision-making is critical. While such models are effective for detecting specific objects, they often fall short in evaluating the broader context of compliance that depends on detailed, site-specific procedures and evolving safety protocols. Safety on the sites often goes beyond the mere presence of PPE, encompassing a range of complex, site-specific protocols and guidelines. Compliance is not just about whether the correct safety equipment is being worn, but also about how those protocols are being followed. For example, workers might be wearing the correct gear, but still fail to adhere to safety guidelines regarding the proper usage of equipment, the maintenance of safe distances from hazardous areas, or the adherence to site-specific rules for handling dangerous materials.

This introduces a challenge for traditional object detection models: they typically focus on recognizing objects in isolation, rather than understanding the broader environment in which these objects are used. This solution addresses that gap by combining video analysis through Vision Language Models (VLMs) ([Bordes et al, 2024](#)) with SOP-driven reasoning.

The conceptual difference between YOLO and VLMs lies in their core paradigms and capabilities. YOLO is primarily an object detection framework designed to perform real-time, high-performance localization and classification of visual objects in images or video streams. It operates on the principle of mapping image pixels to predefined classes in a single pass, focusing on pixel-level object recognition and bounding box predictions. In contrast, VLMs extend beyond traditional image understanding by integrating both visual and linguistic information to enable multi-modal reasoning. VLMs are not limited to mere object detection; they facilitate a deeper understanding of the relationship between visual content and natural language queries, allowing for tasks such as zero-shot recognition, caption generation, and contextual understanding of scenes. While YOLO excels in specific, deterministic tasks like object localization in constrained environments, VLMs represent a more flexible, general-purpose approach to vision tasks by enabling the fusion of textual and visual inputs to infer complex, context-dependent meanings, making them suitable for more dynamic and context-rich applications.

Our solution leverages this advancement in computer vision technologies. Each frame is interpreted using a multimodal AI agent prompted with SOP context, enabling situational understanding beyond object presence. For instance, if a worker enters a restricted zone without proper clearance, even while wearing required PPE, the system correlates this behavior with SOP directives, logs the non-compliant event with a timestamp and contextual description and triggers an automated email notification to designated personnel for rapid response.

Unlike conventional object detection methods like YOLO that focus on specific visual classes, this solution dynamically interprets rig operations based on procedural rules defined in SOPs. It enables the same model to check for multiple forms of compliance across various activities without retraining or predefining all possible violation classes. By combining document parsing, vision analysis, and generative reasoning, the system bridges the gap between static SOPs and dynamic field activity. The findings validate the feasibility of using generative multimodal AI for enhancing operational safety with fidelity, traceability, and scalability.

Paradigm shift in Computer Vision

The landscape of computer vision is undergoing a transformative shift, driven by the emergence of VLMs. [Radford et al, 2021](#) introduces Contrastive Language–Image Pre-training (CLIP) method which shows advancement of vision model from traditional computer vision models which were heavily dependent on large amounts of labelled data and tailored for narrow, task-specific applications. The VLMs can leverage self-supervised and zero-shot learning capabilities, enabling them to understand and interpret images with minimal or no task-specific fine-tuning.

One of the most significant changes brought about by VLMs is their multimodal nature, allowing them to process and integrate both visual and textual information. This leads to a deeper understanding of context and semantics, far beyond what was possible with traditional computer vision architectures.

In this solution, we adopt a multimodal approach to integrate Standard Operating Procedures (SOPs) into our use cases, particularly for video analysis on oil rigs. Our experimental setup incorporates sensor data alongside video feeds to enhance the robustness of the analysis. The availability of off-the-shelf Vision-Language Models (VLMs) significantly accelerates the development process, eliminating the need to build custom models from scratch for each new application.

Methodology

This section presents the architecture and processing flow employed in the development of a hybrid AI-based compliance monitoring system for oil rig operations. The methodology is divided into cloud-based SOP contextualization, edge-based video analysis and compliance evaluation, and cloud-side post-analysis for insight generation.

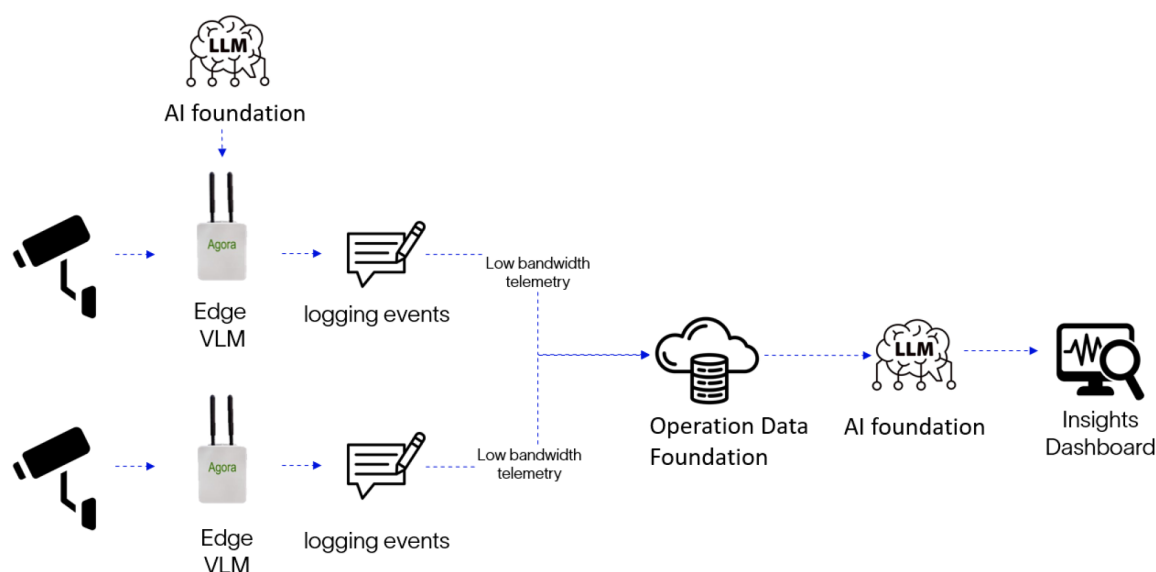


Figure 1—Architecture of the solution

Cloud-Based SOP Contextualization

The system workflow begins with the ingestion of a formal Standard Operating Procedure (SOP) document in the cloud environment. These documents, typically authored by internal safety and operations teams, detail organizational safety protocols, compliance guidelines, and procedural mandates. Upon upload, the SOP document (Fig. 2) is processed through a layout-aware text extraction pipeline hosted in the cloud. This pipeline is capable of parsing complex formatting, including hierarchical headings, tabular data, and safety checklists, thereby enabling comprehensive retrieval of compliance-relevant content.

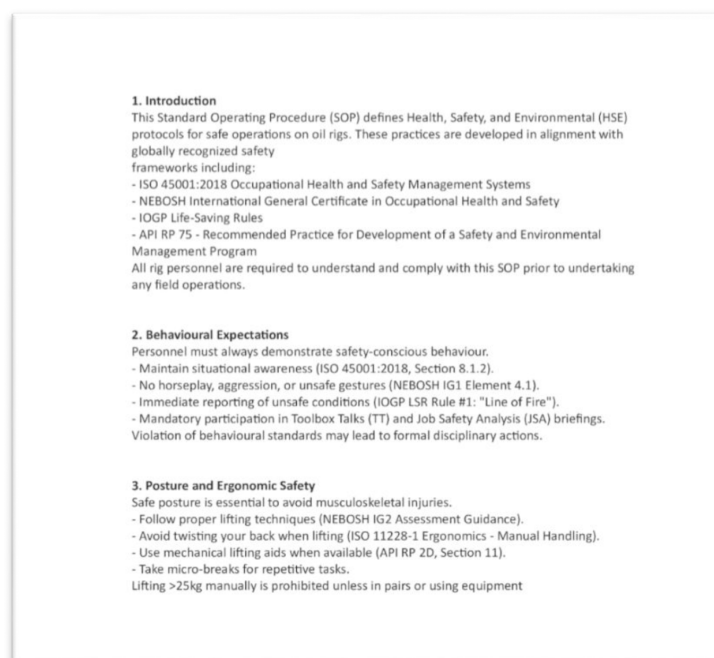


Figure 2—SOP in PDF format

Following extraction, the SOP text is converted into a structured representation of safety rules and operational constraints. A set of context-specific prompts is then automatically generated from this content (Fig.3). These prompts are designed to serve as dynamic inputs to multimodal video analysis agents that

operate downstream. This transformation of static documentation into an actionable knowledge base ensures that the analysis of operational activity is continuously aligned with the current procedural expectations. Once prepared, this contextual information is securely transmitted to the edge device, where evaluation is performed.



Figure 3—Automated context extraction from SOP

Edge-Based Video Analysis and Compliance Monitoring

In the second phase, video feeds acquired from rig-mounted surveillance cameras is fed to the edge device located on-site. This device, equipped with GPU acceleration and optimized for low-latency inference, forms the computational core for localized compliance monitoring. To maintain responsiveness while minimizing computational load, video streams are sampled at a fixed interval. Each frame is independently analysed without assuming temporal continuity.

For real-time processing, we deploy edge AI using compact and efficient hardware like the Nvidia Jetson, which allows us to process camera feeds directly at the source. This setup minimizes latency and reduces reliance on high-bandwidth cloud communication. VLMs, including pre-trained models such as Gemma3 (with 12 billion parameters), are executed on these edge devices, achieving inference speeds of up to 15 tokens per second while maintaining acceptable accuracy levels. The analysis of sampled frames is performed using a multimodal AI engine deployed on the edge device. This engine receives, as part of its prompt, the context extracted from the SOP document during the cloud phase. It evaluates each frame against the organizational safety expectations, assessing whether the visual scene adheres to defined compliance criteria. The agent outputs a structured result per frame, including the compliance status (e.g., Compliant / Non-Compliant), the type of any observed violation (e.g., missing personal protective equipment, unsafe proximity to heavy equipment), and an explanatory rationale rooted in the SOP guidance. By performing

these computations on the edge, the system achieves high-speed, on-site compliance evaluation with minimal reliance on external connectivity.

Cloud-Based Insight Generation from Edge Logs

Upon completion of video analysis, structured logs are generated on the edge device containing detailed metadata on both compliant and non-compliant observations. These logs include timestamps, violation descriptions, and references to the relevant SOP sections. The logs are periodically synchronized back to the cloud for centralized storage and downstream processing. In the cloud, these logs become the foundation for advanced insight generation. Having the detailed as commentary of analysis in the form of sequential observations, aggregated analyses are conducted to uncover trends in compliance behaviour over time and across operational contexts. The system supports the generation of statistical summaries (Fig. 4), visual dashboards, and narrative reports that can be consumed by HSE officers and operational managers. Furthermore, this logged information can be integrated with external datasets, such as equipment telemetry, climate data, and sensor measurements, to enable more holistic safety assessments. For instance, the system can be queried to evaluate whether non-compliance rates correlate with environmental conditions or to determine the safety profile of a facility under specific operational loads. By combining document understanding, edge-based visual intelligence, and cloud-scale analytics, the proposed methodology offers a scalable and contextually grounded framework for continuous HSE compliance monitoring.

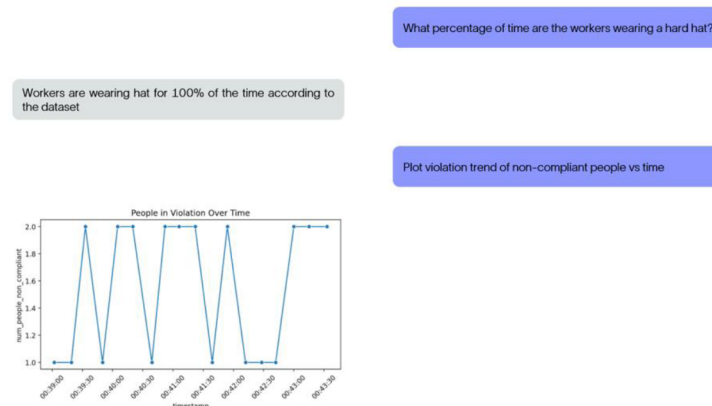


Figure 4—Insights from the logged events

Quantitative results

To assess the effectiveness of our proposed workflow, we conducted a comparative quantitative analysis against a fine-tuned YOLOv11-based PPE detection model. The YOLOv11 model was fine-tuned using a domain-specific dataset consisting of approximately 10,000 annotated images collected from operational oil rigs and workshop environments. The dataset included the following classes: helmet, no_helmet, gloves, no_gloves, person_with_coveralls, person_without_coveralls, shoe, no_shoe, vest, and no_vest. Fine-tuning was performed on a T4 GPU to optimize the model's performance for PPE detection in these industrial environments.

For this evaluation, we focused on two critical safety gear components: helmet detection and the identification of small-scale objects, particularly gloves, where we observed a notable performance degradation in YOLOv11 relative to other PPE classes. Specifically, YOLOv11 struggled with detecting gloves under conditions of partial occlusion and varying visibility, highlighting limitations in its ability to detect smaller or less visible objects compared to larger, more prominent ones like helmets.

In our solution, for the VLM analysis, we enriched the prompt with domain-specific SOP context (as described in section Methodology- Cloud-Based SOP Contextualization) and applied one-shot learning by

providing exemplar references to help the model infer gloves and helmets under varying visibility and occlusion scenarios. For experimentation, we tested a more advanced VLM, Gemini 2.0 Flash, which can perform more complex inference tasks.

To ensure the robustness and reliability of our comparison, both YOLOv11 and the VLM-based solution were evaluated on two distinct sets of previously unseen images. This evaluation allowed us to measure the models’ generalization capabilities and determine the overall effectiveness of the VLM approach relative to the fine-tuned YOLOv11 model in real-world industrial environments.

Example 1

Description of scene: This scene features two rig workers actively engaged in operations on a drilling rig floor. Both individuals wear hard hats, suggesting baseline compliance with head protection protocols. However, machinery partially obscures the view of their bodies, introducing significant visual occlusion. One of the workers appears to be wearing gloves on both hands, while the other worker seems to be wearing a glove only on one hand; the visibility of the other hand is highly limited due to occlusion.

We evaluated 10 frames from this scene. [Table 1](#) summarizes the number of frames (out of 10) in which each model produced correct detections. P1 refers to person 1 and P2 refers to person 2 in [Table 1](#).

Table 1—Comparative study between Fine tuned YOLO V11 model and Multimodal VLM

Model	P1_Helmet	P1_Gloves	P2_Helmet	P2_Gloves
Fine tuned YOLO V11	10	10	10	0
Multimodal VLM	10	10	10	7

Example 2

Description of scene: This scenario involves a single rig worker operating near red-colored drilling equipment. The worker is equipped with a hard hat and red gloves. A notable challenge in this scene arises at some frames due to color camouflage: the red gloves visually blend with the red machinery, resulting in low contrast between the gloves and the background. This introduces a significant difficulty for object detection systems, particularly those reliant on pixel-level feature contrasts.

Across 10 frames from this scene, detection accuracy is summarized in [Table 2](#).

Table 2—Comparative study between fine tuned YOLO V11 model and Multimodal VLM

Model	Helmet	Gloves
Fine tuned YOLO V11	10	4
Multimodal solution	10	8

The qualitative evaluation demonstrates that while the fine-tuned YOLOv11 model performs reliably in detecting helmets, benefiting from their distinct shape and high visual contrast, it exhibits significant limitations in identifying gloves, particularly under challenging conditions such as occlusion or color blending with the background. In contrast, the proposed VLM-based workflow, enhanced with SOP context and one-shot exemplars, shows superior robustness in such edge cases. Notably, the VLM was able to infer glove presence even when visibility was partial or contrast was low, indicating its ability to reason beyond explicit visual cues. These findings suggest that integrating domain knowledge and visual-linguistic reasoning into PPE compliance workflows can substantially improve detection accuracy in complex real-world industrial environments, especially for smaller or visually ambiguous safety equipment like gloves.

Case Studies

Legacy systems are often constrained by lengthy development cycles and limited configurability. In contrast, our solution is designed for agility, featuring shorter development timelines and modular, configurable architectures. These are tailored to leverage the full potential of VLMs, allowing for faster adaptation to new use cases with minimal overhead.

Our proposed workflow distinguishes itself from many current state-of-the-art approaches by offering significant advantages in terms of reduced training requirements, minimized data dependency, and accelerated deployment cycles.

Objective 1: Qualitative Assessment of SOP Compliance Monitoring

The proposed multimodal AI-driven framework was tested across multiple safety-critical scenarios to evaluate its capability in identifying SOP violations using vision-language models (VLMs). Below, we present two case studies that illustrate the system's ability to detect and contextualize non-compliant activities in real-world rig environments.

Case Study 1: Detection of Improper Footing on Drill Pipes. *Objective:* Evaluate the system's ability to identify unsafe behavior based on rig floor SOPs.

Context: The drilling rig floor is a high-risk zone where the presence of heavy machinery, rotating equipment, and stacked drill pipes requires strict adherence to safety procedures.

Observation: In one monitored scenario, the system detected two workers operating on the rig floor. The AI flagged a safety violation involving one worker standing on a horizontally laid drill pipe. This behavior, while potentially overlooked in manual reviews, poses a significant safety risk due to the instability of the pipe and its potential to roll or shift underfoot.

timestamp	result
29-07-2025 00:39	— Compliance Status: Non-Compliant Issues Found: Workers standing on pipes, potentially unstable surfaces. Section of SOP in non-compliance: Safe Work Practices Number of people non-compliant in the frame: 1 —

Analysis: The system correctly interpreted the contextual elements of the scene i.e horizontal pipe, the worker's position, and surrounding equipment, and matched them against the textual SOPs. The incident corresponds to a direct violation outlined in the rig safety SOPs, which explicitly prohibit standing on loose or unsecured pipes to prevent trip and crush hazards.

Implication: This case demonstrates the model's ability to identify subtle but critical deviations from SOPs without needing rule-based programming or hardcoded image templates. The insight generated could support real-time alerts or post-operation safety audit

Case Study 2: Simulated Fall-from-Height Detection on Rig. *Objective:* Evaluate the system's effectiveness in detecting fall hazards through a controlled simulation aligned with "Working at Heights" SOPs.

Context: Fall-from-height incidents remain one of the most critical safety concerns in drilling operations. To assess the proposed framework's ability to detect such hazards, a controlled simulation was carried out involving a dummy representing a worker positioned at an elevated section of the rig structure.

Observation: During the simulation, the dummy was placed to mimic a technician performing tasks at height. The setup included typical rig equipment and environmental context. A fall was then triggered, causing the dummy to drop from the platform to the rig floor below. The AI system monitored the scene in real-time using CCTV feeds.

timestamp	result
06-08-2025 01:19	— Compliance Status: Non-Compliant Issues Found: One individual is observed working at height (greater than 6 feet) on the structure, and it is not evident from the image if they are adequately protected by fall protection systems such as guardrails, a fall restraint system, or a personal fall arrest system (PFAS). There is no visible indication of the worker being tied-off. Section of SOP in non compliance: Section 2. Scope; Section 3, Item 1; Section 6.A; Section 7.1.D Number of people non-compliant in the frame: 1 —
06-08-2025 01:19	— Compliance Status: Non-Compliant Issues Found: An individual is observed working at height on the derrick structure, which appears to be greater than six feet from the ground, without any visible fall protection system (e.g., guardrails, fall restraint systems, or personal fall arrest systems). Section of SOP in non compliance: Focus Area: General Fall Protection Triggers and Requirements – "All employees and contractors performing work on behalf of USPL where there is a potential to fall six (6) feet or more must be protected." (OSHA 29 CFR 1926 Subpart M) (Section 2. Scope, Section 3, Item 1; Section 6.A); "Fall protection (guardrail systems, fall restraint systems, personal fall arrest systems, or safety net systems) is required anytime an individual is exposed to a fall of 6 feet or greater while performing work." (Company Policy; OSHA regulations) (Section 3, Item 1; Section 6.A); Focus Area: Fall Protection Systems (PFAS, Fall Restraint, Guardrails, Safety Nets, Warning Lines, Positioning Devices) – "Personnel using a PFAS shall be tied-off at all times when working > 6 feet from the ground or next lower level." (Company Policy) (Section 7.1.D). Number of people non-compliant in the frame: 1 —

Analysis: The system successfully detected the fall event within a few seconds. It identified the upright posture of the dummy before the fall, the absence of a visible safety harness or fall arrest system, and the rapid downward motion as key indicators of non-compliance with "Working at Heights" SOPs. The fall was interpreted as a critical safety violation, and the system generated a high-severity alert.

Implication: This case demonstrates the model's capacity for recognizing fall events with minimal supervision, using spatiotemporal cues and contextual understanding of rig environments. The success of this simulation highlights the framework's potential for real-time incident detection and proactive safety intervention, especially in environments where manual monitoring may be delayed or inconsistent.

Objective 2: Event-Based Summarization in Warehouse Operations

The proposed solution extends beyond real-time compliance monitoring to support **event-based video summarization**. This capability enables the system to extract and sequence key activities from continuous video footage using a configurable sampling rate or a defined temporal resolution. Such summarization helps safety teams, supervisors, and auditors rapidly interpret critical events without reviewing the entire footage manually.

Use Case: Automated Summarization in Warehouse Environment. *Context:* A warehouse environment was selected as a test scenario to evaluate the framework's ability to summarize activity in a busy, dynamic setting. The video captured typical operational scenes involving workers and a moving forklift.

Result: The AI system generated structured, step-wise textual summaries of the video events is shown in [Figure 5](#).

Analysis: These summaries are not separate observations but represent different segments of the same video, spaced across continuous timestamps. As the scene unfolds, the AI dynamically updates the summary with new contextual information, demonstrating temporal awareness and evolving scene understanding.

Objective 3: Beyond safety and operational efficiency, our solution presents a significant opportunity for enhancing environmental sustainability like in the context of flare monitoring.

Implication: This demonstrates the system's ability to maintain temporal continuity in video understanding. By summarizing evolving events over time, it supports proactive situational awareness and enables rapid review of long-duration footage. Such capabilities are especially valuable in large facilities, where manual review of surveillance feeds is resource-intensive.

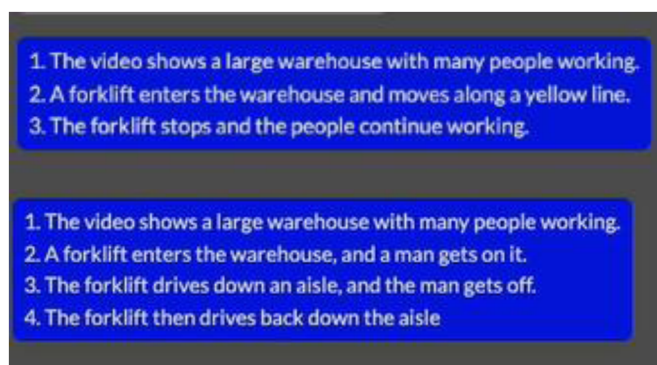


Figure 5—Scene summarisation of a warehouse operation

Objective 3: Visual Grounding for Flare Monitoring

Beyond safety and operational efficiency, the proposed solution explores new frontiers in environmental sustainability, particularly in the context of flare monitoring. As a proof-of-concept, we evaluate our solution to localize and describe flaring events through bounding box outputs on visual data.

Use Case: Bounding Box Detection of Flare Activity. *Context:* Flaring is a critical environmental concern in oil and gas operations. Accurate and real-time detection of flaring incidents is essential for emission control and regulatory compliance. To explore the applicability for this task, a conceptual test was performed using several images depicting an industrial scene with a flare stack.

Result: The VLM successfully identified the flare within the image and generated a bounding box around the flame region (Figure 6). The detection demonstrates the model's ability to visually ground key elements of environmental interest, even in stylized or non-photorealistic data.



Figure 6—Flare monitoring bounding box on a cartoon image

Analysis: Although the current demonstration uses a cartoon image, the result showcases scene understanding and localization capabilities. The model was able to semantically associate the flare concept with the visual representation, suggesting feasibility for real-world extension using CCTV or drone footage.

This bounding box-based visual grounding serves as an important step toward automating flare detection pipelines. Combined with time-stamped footage and integration into compliance dashboards, it can enhance emission monitoring workflows.

Implication: This early-stage validation highlights the scalability and adaptability of the proposed framework in environmental monitoring applications. By leveraging VLMs, the system could eventually support real-time flare event tracking without requiring specialized hardware—using only existing visual infrastructure.

Implication: This early-stage validation highlights the scalability and adaptability of the proposed framework in environmental monitoring applications.

With the ease to develop the solution fast, the scope of implementation of our solution is vast. Given the rapid development cycle and inherent adaptability of our solution, its scope of implementation across various industrial applications is demonstrably vast. This ease of implementation unlocks a broad spectrum of uses, ranging from ensuring stringent SOP compliance, through intelligent video summarization for operational insights, to supporting vital sustainability objectives such as comprehensive flare monitoring. These diverse applications collectively demonstrate the solution's potential to drive efficiency and safety across industrial environments.

Conclusion

This research presents a novel AI-powered compliance monitoring solution that fundamentally redefines safety and compliance assurance in high-risk industrial environments, specifically drilling and production. Our system integrates cloud-based SOP contextualization with real-time edge-based video processing to enable continuous, intelligent monitoring of frontline operations. This architecture transitions from conventional event-based reporting to a comprehensive full-scene understanding, thereby enhancing the granularity of safety insights and facilitating a proactive, rather than reactive, approach to risk mitigation. The culmination of this framework is a next-generation safety index, a dynamic, near real-time indicator that quantifies operational safety states and fosters superior compliance adherence within organizational workflows.

The system's operational efficacy is underpinned by its deployment on lightweight edge devices, such as NVIDIA Jetson modules, integrated within robust field-deployable platforms like Agora. These edge devices facilitate on-device processing of high-resolution camera feeds utilizing VLMs. These VLMs are specifically designed for localized, SOP-aligned inference, ensuring immediate identification of non-compliant behaviors or conditions. Furthermore, this low-power edge setup, with an approximate consumption of 75W in contrast to typical GPU setups demanding ~1000W, significantly reduces the energy footprint associated with AI deployments, thereby directly contributing to corporate emissions reduction objectives. The real-time detection and response capabilities for incidents such as spills, unsafe work practices, or Personal Protective Equipment (PPE) violations further support emission reductions by mitigating incident escalation and improving critical reaction times to safety-critical conditions.

On the cloud-side, all localized frame-level assessments are transmitted as structured logs, establishing high-resolution behavioral records. These logs undergo advanced enrichment via Large Language Models (LLMs) and are meticulously integrated with diverse auxiliary data streams, including sensor readings, environmental metrics, and operational metadata. This comprehensive data integration enables sophisticated downstream analytics, encompassing automated pattern detection, comprehensive safety summarization, and seamless integration with advanced reporting frameworks. Such capabilities ultimately lead to optimized resource utilization, minimization of waste, and the identification of targeted coaching opportunities for continuous workforce development. This holistic system represents a paradigm shift from static safety documentation to a dynamic, intelligent compliance ecosystem, generating substantial long-term value for both operational entities and safety stakeholders.

In terms of market and technical readiness, this solution is currently in the Proof of Concept (POC) stage, with a focus on refining scalability and ensuring optimal hardware for edge deployment. While the core functionality has been validated in initial tests, further work is required to tailor the system for large-scale industrial environments, including the optimization of hardware infrastructure and the deployment of energy-efficient, high-performance edge devices that support real-time processing. Ongoing field testing is crucial to assessing the solution's robustness under dynamic operational conditions and to fine-tune the integration of real-time data streams from sensors, video, and environmental metrics. The solution is

progressing rapidly, with additional emphasis on scalability and hardware selection to ensure that the system can handle the complexities of large sites and high-volume data without compromising performance. As the system matures, its deployment will provide substantial improvements in operational safety, efficiency, and compliance adherence, positioning it as a transformative tool for modern industrial environments.

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